

Douglas Hanley

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EMPLOYMENT

Founding Engineer, Compendium Labs, 2024 —

Associate Professor, Department of Economics, University of Pittsburgh,
2023 — 2025

Assistant Professor, Department of Economics, University of Pittsburgh,
2014 — 2023

Visiting Scholar, Federal Reserve Bank of St. Louis, 2017 (short term)

EDUCATION

Ph.D., Economics, University of Pennsylvania, 2014

B.S., Economics, California Institute of Technology, 2006

PUBLICATIONS

Optimal Taxation and R&D Policies (with Ufuk Akcigit and Stefanie Stantcheva), *Econometrica*, 2022

High-speed Railways and Collaborative Innovation (with Jiancheng Li and Mingqin Wu), *Regional Science and Urban Economics*, 2022

Back to Basics: Basic Research Spillovers, Innovation Policy and Growth (with Ufuk Akcigit and Nicolas Serrano-Velarde), *Review of Economic Studies*, 2021

Cited in the [Economic Report of the President 2016](#)

Assessing the Rate of Replication in Economics (with James Berry, Lucas Coffman, Rania Gihleb, and Alistair Wilson), *American Economic Review (Papers and Proceedings)*, 2017

Transition to Clean Technology (with Daron Acemoglu, Ufuk Akcigit, and William Kerr), *Journal of Political Economy*, 2016

WORKING PAPERS

Unleashing the Dragon: The Case for Patent Reform in China (with Lee Branstetter and Huiyan Zhang)

Science is Shaped by Wikipedia: Evidence from a Randomized Control Trial (with Neil Thompson)

From Patents to Products: Innovation, Product Creation, and Firm Growth (with David Argente, Salomé Baslandze, and Sara Moreira)

Economics and Epidemics: Evidence from an Estimated Spatial Econ-SIR Model (with Mark Bognanni, Daniel Kolliner, and Kurt Mitman)

Innovation and Patent Policy with Interdependent Technology

RESEARCH IN PROGRESS	Measuring Patent Novelty Using Text Embeddings Spatial Clustering of Firms: A Machine Learning Approach (with Chengying Luo and Mingqin Wu) Theory and Evidence on the Relationship Between Corruption and Innovation (with Mingqin Wu) International Competition in the Race to Clean Technology (with Chengying Luo) Yellow Fever and the Growth of American Cities (with Sean McCoy and Werner Troesken)
TEACHING EXPERIENCE	Graduate: Computational Methods in Economics, Advanced Economic Growth, Advanced Macroeconomic Theory I/II Undergraduate: Macroeconomics Proseminar, Economic Growth Elective, Intermediate Macroeconomics, Intermediate Microeconomics (Penn)
SEMINARS	2024: Miami University, Rice University 2023: Federal Reserve Board, Federal Reserve Bank of Dallas, NBER Productivity Seminar (Virtual) 2019: Federal Reserve Bank of Cleveland, Seoul National University, South China Normal University (Guangzhou) 2018: World Bank, Einaudi Institute for Economics and Finance (Rome), South China Normal University (Guangzhou) 2015-2017: Seoul National University, U of Hong Kong, Sogang U (Seoul), IIES (Stockholm) 2014: U of Toronto, USC (Marshall), U of Pittsburgh, U of North Carolina (Chapel Hill), CREI (Barcelona)

CONFERENCES

2024: Banff Innovation, Productivity and Growth Conference

2023: NBER Productivity, Innovation, and Entrepreneurship Meeting (Cambridge), Junior Innovation Economics Conference (Harvard Business School)

2022: 9th ZEW/MaCCI Conference on the Economics of Innovation and Patenting (Frankfurt)

2021: Souther Economic Association Annual Meeting (virtual, Houston), Taiwan Symposium on Innovation Economics and Entrepreneurship (virtual, National Tsing Hua University)

2019: SED Annual Meeting (St. Louis), Korea Economic Review International Conference (Seoul), Econometrics Society China (Jinan University, Guangzhou), Roundtable for Engineering Entrepreneurship Research (Georgia Tech, discussant)

2018: Computational Social Sciences Retreat (Pitt), Econometrics Society Asia Meetings (Seoul), 4th Workshop on Chinese Politics and Society (Shanghai)

2017: Workshop on Innovation and Entrepreneurship (ISET-TSU, Tbilisi), Carnegie-Rochester-NYU Conference on Public Policy (CMU, discussant)

2016: Chinese Economists Society China Meeting (PHBS, Shenzhen)

2015: Firm and Technology Dynamics: A Global Perspective (Nottingham), Barcelona GSE Summer Forum, SED Annual Meeting (Warsaw), Mini Conference on Economic Growth (Chicago)

2010-2014: Penn Money Macro Workshop, SED Annual Meeting, Midwest Macroeconomic Meetings, Penn Money Macro Club, International Industrial Organization Conference

STUDENTS SUPERVIZED

David Agorastos (Ph.D. 2024): Research Financial Economist, Office of the Comptroller of the Currency

Ying-Kai Huang (Ph.D. 2021): Applied Research Scientist - Economist, Snap Inc

THESIS COMMITTEES	Mikhail Zavarzin (Ph.D. 2025): Post-doctoral Researcher, HEC Lausanne Hokyu Song (Ph.D. 2024): Associate Research Fellow, Korea Development Institute Hyungjin Kim (Ph.D. 2024): Associate Research Fellow, Korea Development Institute Steven Lann (Ph.D. 2024): Data Science Fellow, Allegheny County Department of Human Services Lucy Wang (Ph.D. 2023, Katz): Postdoctoral Researcher, University of Cambridge Domonkos Vamosy (Ph.D. 2020): Economist, Amazon Web Services Tianyi Wang (Ph.D. 2020): Postdoctoral Research Associate, Princeton Jiyeon Kim (Ph.D. 2020): Economist, Korea Development Institute Chengying Luo (Ph.D. 2019): Senior Data Scientist, Instacart Hao Feng (Ph.D. 2018): Assistant Professor, Shanghai University of International Business and Economics Richard Svoboda (Ph.D. 2017): Senior Economist, Amazon
REFEREING SERVICE	American Economic Review, Econometrica, Review of Economic Studies, Journal of Political Economy, Journal of Monetary Economics, European Economic Review, American Economic Journal: Macroeconomics, Journal of Public Economics, International Economic Review, Macroeconomic Dynamics, Review of Economic Dynamics, Journal of Comparative Economics, Journal of the European Economic Association, Journal of Mathematical Economics, Journal of Development Economics, Review of Economics and Statistics, Journal of Economic Theory, RAND Journal of Economics, The Economic Journal
OTHER EMPLOYMENT	Research Assistant, MIT, 2010-2013 Research Assistant, Caltech, 2005-2006 Research Assistant, CITA, Summer 2004 Software Developer, Cyplex Inc, 2006-2007 Software Developer, Lycoris Corp, Summer 2003 Software Developer, KDE Group, 2001-2002
